Semi-Supervised Sentiment Analysis of Farsi (Dari) Social Media Text

Azizullah Massomy
a12244794@univie.ac.at

Project Report for Practical 2

Supervised by: ao. Univ.-Prof. Mag. Mag. Dr.Werner Winiwarter

March 26, 2026, Vienna

Contents

1 Abstract

In my project, I explore a semi-supervised emotion classification study on Persian (Dari) social media text. Building on Phase 1, where a manually annotated dataset of 4,000 comments and messages were created and used to fine-tune ParsBERT, the second phase focuses on scaling the dataset and improving model performance under severe class imbalance. Using the Phase 1 model as a labeling agent, approximately 390,000 previously unlabeled comments were automatically annotated and employed for incremental fine-tuning.

A central challenge addressed in this phase is the extreme imbalance of low-frequency emotions, particularly *Fear*. To systematically analyze its impact, multiple experimental configurations were evaluated, including an eight-class setting, a seven-class ablation excluding *Fear*, and a reintroduced eight-class model enhanced with targeted data augmentation. While the initial eight-class model achieved high overall accuracy, the *Fear* category remained poorly learned. Removing this class improved macro-level performance, confirming its destabilizing effect. Reintroducing *Fear* through culturally informed augmentation resulted in the best overall performance, achieving an accuracy of 88.87% and a substantial improvement in *Fear* classification.

My results show that semi-supervised learning, combined with targeted augmentation strategies, can effectively scale emotion classification for low-resource languages such as Dari. At the same time, qualitative analysis reveals persistent confusion among semantically similar negative emotions, highlighting directions for future refinement. Several implementation-related challenges, such as parameter tuning and result stability, influenced the final experimental setup.
Keywords: Emotion Classification, Persian NLP, Dari Language, Semi-Supervised Learning, Class Imbalance, Data Augmentation

2 Introduction

2.1 Context and Motivation

Emotion classification is a fundamental task in sentiment analysis, with applications ranging from public opinion monitoring to crisis detection and social media analytics. While substantial progress has been made for high-resource languages, emotion analysis for low-resource languages such as Persian (Dari) remains challenging due to limited labeled data, dialectal variation, and culturally specific emotional expressions.

In Phase 1 of my project, i addressed the lack of annotated resources by manually labeling a dataset of 4,000 Dari social media comments and fine-tuning ParsBERT for eight emotion categories. Although this effort established a working baseline, the resulting model achieved only moderate performance, with an overall accuracy of 53.89%. More importantly, the analysis revealed two major limitations: the scalability constraints of manual annotation and the severe class imbalance affecting low-frequency emotions, particularly *Fear* and *Disgust*. These limitations motivated design of my project second phase focused on scaling and robustness.

2.2 My Learning Journey

When I started Phase 1, I underestimated how challenging emotion classification would be for Dari. As a native speaker, I assumed I could easily distinguish emotions, but translating that intuition into consistent labels proved difficult. The 53.89% accuracy of my Phase 1 model was humbling but motivated me to explore more sophisticated approaches in Phase 2.

I chose semi-supervised learning not just because it's efficient, but because I wanted to test whether my initial manual annotations—despite being limited—could bootstrap a larger system. This reflects my growing understanding of practical NLP: starting small, validating carefully, then scaling.

2.3 Phase 2 Objectives

Phase 2 builds directly upon the outcomes of Phase 1 and aims to overcome its identified shortcomings through a semi-supervised learning strategy. Instead of relying solely on additional manual annotation, the Phase 1 model is leveraged as an automatic labeling agent to expand the dataset substantially. This approach allows the model to learn from a much larger corpus while preserving the label structure defined in the initial phase.

The objectives of Phase 2 are threefold:

1. **Dataset scaling**: Expand the labeled dataset from 4,000 manually annotated samples to approximately 390,000 samples through automatic labeling.
2. **Imbalance analysis**: Systematically investigate the impact of extreme class imbalance by comparing eight-class and seven-class model configurations.
3. **Targeted augmentation**: Improve the learning of the low-resource *Fear* class through culturally informed data augmentation and re-evaluation.

2.4 Research Questions

Based on these objectives, this phase addresses the following research questions:

- Can semi-supervised learning effectively scale emotion classification for Persian (Dari) without degrading label quality?
- How does extreme class imbalance influence macro-level performance in multi-class emotion classification?
- To what extent can targeted data augmentation mitigate the weaknesses of low-frequency emotion classes?

2.5 Report Structure

The remainder of this report is organized as follows. Section 2 reviews related work on emotion classification and semi-supervised learning for low-resource languages. Section 3 presents the methodology, including the semi-supervised pipeline and data augmentation strategy. Section 4 describes the experimental setup. Section 5 reports quantitative and qualitative results, which are further analyzed in Section 6 through discussion and interpretation. Finally, Section 7 concludes the report and outlines directions for future work.

3 Related Work

This section situates the present work within existing research on emotion analysis for Persian and low-resource languages, with particular emphasis on challenges relevant to Phase 2 of my project.

Emotion and sentiment analysis has been extensively studied for high-resource languages, particularly English. However, for Persian and its regional variants such as Dari, research remains limited, especially for fine-grained emotion classification under real-world data imbalance. This section reviews prior work most relevant to this study, focusing on three themes: (1) Persian emotion and sentiment analysis, (2) class imbalance and rare emotion modeling, and (3) data augmentation strategies for low-resource emotions.

3.1 Emotion and Sentiment Analysis for Persian

Early work on Persian sentiment analysis primarily focused on polarity classification (positive/negative) using classical machine learning approaches and small curated datasets. With the introduction of transformer-based models, recent studies have demonstrated that pre-trained Persian language models significantly outperform traditional methods. Farahani et al. [1] introduced ParsBERT, a RoBERTa-style model trained on large Persian corpora, which has since become the dominant backbone for Persian NLP tasks, including sentiment and emotion classification.

Several recent studies systematically evaluated transformer models on Persian subjective tasks, including emotion analysis, toxicity detection, and stance classification. A comprehensive survey and benchmark study [2] highlights that, despite strong overall accuracy, Persian emotion models exhibit high instability across datasets and struggle particularly with minority emotions. These findings directly align with the observations in my work, where overall accuracy masked severe weaknesses in low-frequency emotion classes such as *Fear*.

3.2 Fine-Grained Emotion Modeling and Class Imbalance

Fine-grained emotion classification introduces additional challenges compared to binary sentiment analysis, particularly when emotion categories are unevenly distributed. Several studies emphasize that macro-level evaluation metrics (e.g., macro-F1) are essential for meaningful assessment in such settings, as accuracy alone is dominated by majority classes [3].

Recent work on emotion classification across languages shows that rare emotions are consistently under-learned, even when using large transformer models [4]. These studies report systematic confusion among semantically related negative emotions, such as sadness, anger, and fear. Similar confusion patterns were observed in this study, where *Sad* frequently absorbed *Anger* and *Fear* predictions in the absence of sufficient labeled data.

Class weighting and re-sampling are commonly proposed solutions, but multiple studies caution that extreme weighting can lead to unstable training and overconfident predictions [5]. This supports the design choice in this work to rely primarily on data-level interventions rather than aggressive loss re-weighting.

3.3 Data Augmentation for Low-Resource Emotion Classes

Data augmentation has been widely explored as a strategy to mitigate data scarcity and class imbalance in emotion classification. While early work primarily relied on generic text transformations, recent studies have questioned the effectiveness of augmentation when applied without semantic or cultural awareness. A large-scale evaluation by [6] shows that naive augmentation techniques often yield inconsistent improvements and may even degrade performance for fine-grained emotion tasks.

More recent research emphasizes that augmentation benefits are highly class-dependent and are most effective when tailored to specific minority emotions [7]. In the context of low-resource languages, this challenge is further compounded by the fact that emotional expression is closely tied to cultural, social, and political context. Generic paraphrasing or noise-based augmentation often fails to preserve these nuances, limiting its usefulness for emotions such as fear or anxiety.

Studies focusing on culturally grounded emotion modeling highlight the importance of context-aware data generation, particularly in conflict-affected or politically unstable regions [8]. These findings suggest that emotion classifiers trained on such data benefit from augmentation strategies that explicitly encode culturally relevant triggers and expressions, rather than relying solely on surface-level text variation.

In addition to quantitative evaluation, recent work stresses the role of qualitative analysis in understanding the effects of augmentation and class imbalance. Error-focused studies [9, 10] demonstrate that improvements in overall accuracy may obscure persistent confusion among semantically related emotions. This further motivates targeted augmentation approaches that are evaluated not only through aggregate metrics but also through detailed error analysis.

Together, these findings motivate the semi-supervised and data-centric methodology adopted in Phase 2, which is described in the next section.

4 Methodology

This section describes the methodology adopted in Phase 2 of the project. Building directly on the Phase 1 model and preprocessing pipeline, the focus of this phase is on large-scale dataset expansion, systematic handling of class imbalance, and targeted improvement of low-frequency emotion classes. The overall workflow follows a semi-supervised learning paradigm and is summarized in Figure 1.

4.1 Overview of the Semi-Supervised Pipeline

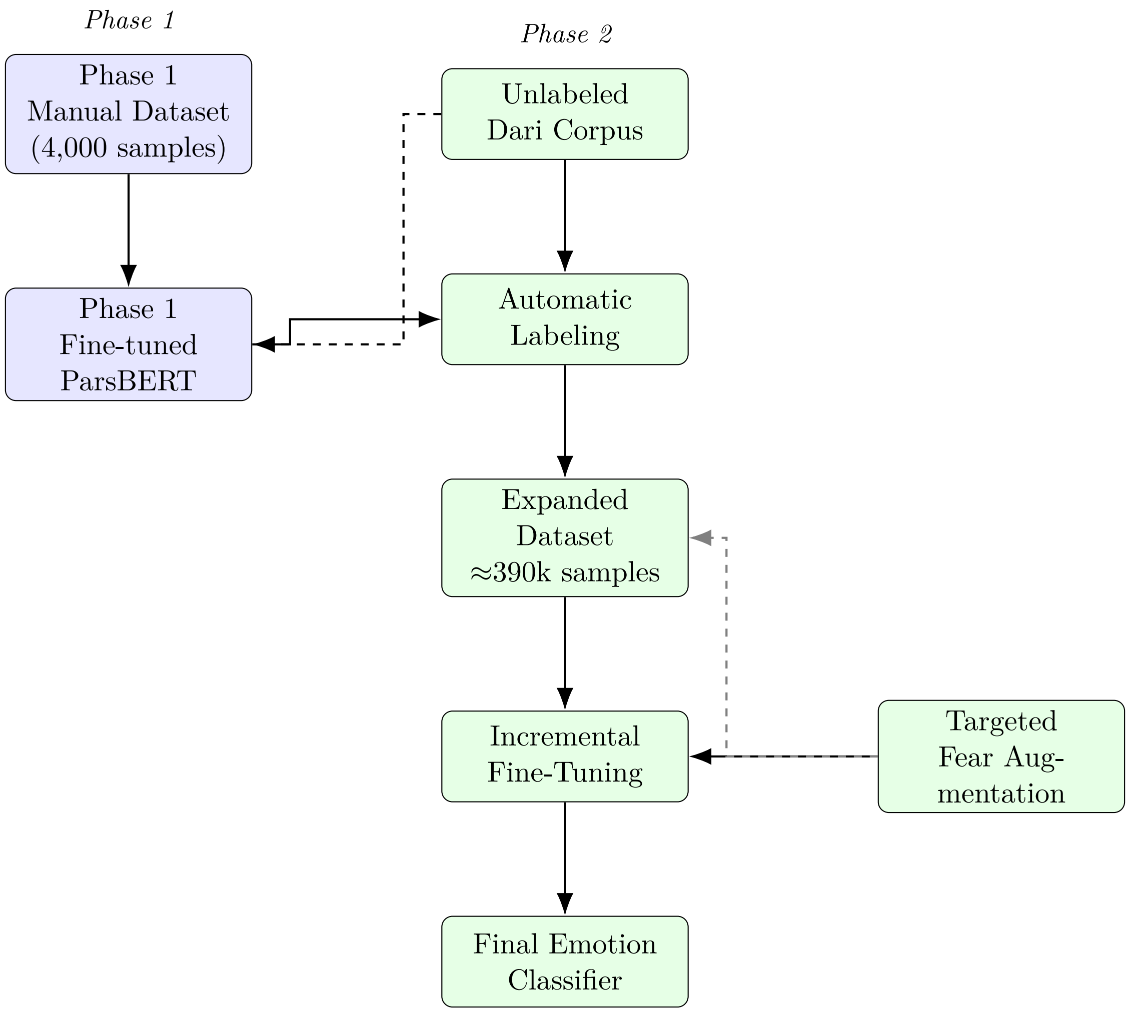

Figure 1: Semi-supervised pipeline used in Phase 2. The Phase 1 model is employed for automatic labeling, followed by incremental fine-tuning and targeted augmentation of the *Fear* class.

Phase 2 employs a self-training strategy in which the fine-tuned ParsBERT model from Phase 1 is used to automatically label a large collection of previously unlabeled Persian (Dari) social media comments. These pseudo-labeled samples are then incorporated into an incremental fine-tuning process to improve model robustness and generalization.

The pipeline consists of four main stages: preprocessing, automatic labeling, incremental fine-tuning, and imbalance-aware refinement.

4.2 Dataset Expansion via Automatic Labeling

The Phase 1 ParsBERT model was applied to the remaining unlabeled portion of the corpus using the same preprocessing steps as in Phase 1, including text normalization, removal of non-Persian characters, and filtering of empty or malformed entries. This ensured consistency between manually labeled and automatically labeled data.

Through my work process, the dataset was expanded from 4,000 manually annotated samples to approximately 390,000 pseudo-labeled samples. A summary of the resulting class distribution is provided in Table 1. As expected, the automatic labeling process amplified existing class imbalance, with emotions such as *Fear* remaining extremely underrepresented.

Emotion	**Original Count**	**Percentage (%)**	**After Augmentation**
Happy	123 222	31.55	123 222
Neutral	82 911	21.23	82 911
Sad	75 277	19.27	75 277
Hope	36 123	9.25	36 123
Disgust	28 469	7.29	28 469
Anger	26 636	6.82	26 636
Surprise	17 956	4.60	17 956
Fear	1097	0.28	10 097
Total	390 691	100.00	400 691

Table 1: Dataset statistics and emotion class distribution after automatic labeling and targeted augmentation.

4.3 Incremental Fine-Tuning Strategy

Rather than training a new model from scratch, incremental fine-tuning was performed starting from the Phase 1 checkpoint. This choice preserves the linguistic and emotional representations learned during manual annotation while allowing the model to adapt to the larger dataset.

The expanded dataset was split into training and validation sets using an 80/20 stratified split to maintain emotion distributions across splits. Training was conducted using the same transformer architecture, with learning rates explored in the range of 1×10^{-5} to 5×10^{-5}. Early stopping with a patience of three epochs was applied to reduce overfitting. Model performance was evaluated using accuracy and macro-F1 score, with macro-F1 serving as the primary metric due to severe class imbalance.

4.4 Handling Class Imbalance

Severe imbalance among emotion classes emerged as a central challenge in Phase 2, particularly for the *Fear* category, which constituted less than one percent of the dataset after automatic labeling. Several complementary strategies were explored to analyze and mitigate this issue.

First, class-weighted loss functions were evaluated to compensate for unequal class frequencies. However, following observations in prior work, aggressive weighting was found to introduce training instability. Second, an ablation experiment was conducted in which the *Fear* class was removed entirely, resulting in a seven-class classification setting. This configuration allowed for isolation of the impact of the rare class on overall performance.

4.5 Targeted Data Augmentation for the Fear Class

To reintroduce the *Fear* class in a controlled manner, a targeted data augmentation strategy was designed. Instead of applying generic text transformations, augmentation focused on generating culturally and contextually plausible expressions of fear in Dari. This approach was informed by prior findings on culturally grounded emotion modeling.

Augmented samples were generated by combining fear-related actors, scenarios, and emotion markers commonly observed in Dari social media discourse. Examples include references to conflict, economic uncertainty, and future-related anxiety. Generated samples were deduplicated and manually inspected for linguistic plausibility before being added to the training set. The final augmented dataset statistics are summarized in Table 1 and visually illustrated in Figure 2.

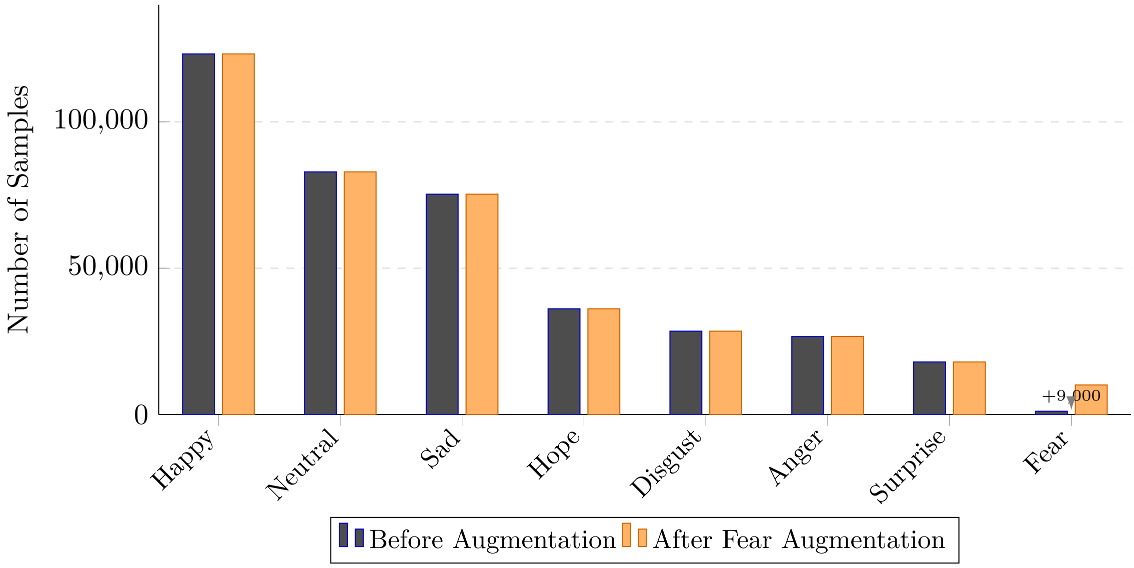

Figure 2: Emotion class distribution before and after targeted augmentation of the *Fear* class. Augmentation increases minority class representation by adding 9,000 synthetic samples without altering majority class proportions.

4.6 Experimental Configurations

Based on the strategies above, three main experimental configurations were evaluated:

Configuration	Description	# Classes
8-Class Baseline	Model trained on automatically labeled dataset with severe class imbalance	8
7-Class Ablation	Model trained after removing the *Fear* category to assess its impact	7
8-Class Augmented	Model trained with targeted *Fear* augmentation to address imbalance	8

Table 2: Summary of experimental configurations evaluated in Phase 2.

These configurations enable a systematic comparison of imbalance effects and the effectiveness of data-level interventions, forming the basis for the quantitative and qualitative analyses presented in the next sections.

5 Experiments

This section describes the experimental setup used to evaluate the proposed semi-supervised emotion classification approach. The focus is on training configurations, data splits, and evaluation metrics, while quantitative and qualitative results are presented in the subsequent section.

5.1 Data Splitting Strategy

For all experimental configurations, the dataset was divided into training and validation subsets using an 80/20 split. Stratified sampling was applied to preserve the original emotion distribution across splits, ensuring that minority classes such as *Fear* were represented in both sets. This strategy was used consistently across all experiments to allow fair comparison between model configurations.

5.2 Training Configuration

All models were fine-tuned using the ParsBERT transformer architecture. Training was performed using the HuggingFace Transformers framework with PyTorch as the backend. The AdamW optimizer was employed, as commonly recommended for transformer-based models.

Learning rates were explored in the range of 1×10^{-5} to 5×10^{-5}, with the final configuration selected based on validation stability. Models were trained for a maximum of 10 epochs, with early stopping applied using a patience of three epochs to prevent overfitting. Mini-batch training was conducted with a batch size selected to balance convergence stability and hardware limitations.

To improve reproducibility, all experiments were conducted using fixed random seeds for data splitting and model initialization.

5.3 Loss Function and Optimization

Cross-entropy loss was used as the primary objective function for all experiments. In selected configurations, class-weighted loss was evaluated to compensate for imbalanced emotion distributions. However, as discussed in the Methodology section, aggressive weighting was found to introduce training instability and was therefore not used as the primary strategy in the final configurations.

Gradient clipping was applied during training to reduce the risk of exploding gradients, particularly when fine-tuning on the large automatically labeled dataset.

5.4 Evaluation Metrics

Model performance was assessed using both accuracy and macro-averaged F1 score. While accuracy provides an overall measure of correctness, it is heavily influenced by majority classes in imbalanced datasets. Macro-F1 was therefore treated as the primary evaluation metric, as it assigns equal importance to each emotion class and better reflects performance on low-frequency emotions.

In addition to aggregate metrics, per-class precision, recall, and F1 scores were computed to support detailed error analysis in later sections.

5.5 Experimental Configurations

Based on the methodology described earlier, three experimental configurations were evaluated:

- **Eight-class model**: Trained on the automatically labeled dataset with all original emotion categories.
- **Seven-class ablation model**: Trained after removing the *Fear* class to isolate the impact of extreme class imbalance.
- **Augmented eight-class model**: Trained with the reintroduced *Fear* class using targeted data augmentation.

These configurations enable systematic analysis of the effects of class imbalance and data-level interventions under otherwise comparable training conditions.

6 Results

This section presents the quantitative and qualitative results obtained from the experimental configurations described in the previous section. Performance is reported using accuracy and macro-averaged F1 score, along with class-level metrics to highlight the behavior of minority emotions. Interpretation and implications of these results are discussed in Section 7.

6.1 Overall Performance Comparison

Table 3 summarizes the performance of all experimental configurations. The results highlight clear differences between the eight-class baseline, the seven-class ablation setting, and the augmented eight-class model.

Model Configuration	**Accuracy (%)**	**Macro-F1**	**Fear F1**
Phase 1 Baseline (4k manual)	53.89	0.5217	–
P2: Eight-class (imbalanced)	85.87	0.7872	0.5212
P2: Seven-class (no Fear)	85.88	0.8240	–
P2: Eight-class (augmented)	88.87	0.8390	0.9462

Table 3: Performance comparison across experimental configurations.

6.2 Impact of Class Imbalance

The eight-class model trained on automatically labeled data achieved high overall accuracy but exhibited reduced macro-F1 performance due to poor learning of low-frequency emotion classes. In particular, the *Fear* category showed substantially lower F1 score compared to majority emotions.

Removing the *Fear* class in the seven-class configuration resulted in an increase in macro-F1 score, while accuracy remained largely unchanged. This configuration provides a useful reference point for assessing the effect of extreme class imbalance on aggregate performance.

6.3 Effect of Targeted Data Augmentation

Reintroducing the *Fear* class through targeted data augmentation led to improvements across all reported metrics. The augmented eight-class model achieved the highest accuracy and macro-F1 score among all configurations. Most notably, the F1 score for the *Fear* class increased substantially, indicating improved learnability after augmentation.

These results demonstrate that data-level interventions can substantially improve minority class performance without degrading majority class accuracy.

6.4 Qualitative Error Analysis

To complement quantitative metrics, a qualitative analysis was conducted on representative validation samples. Table 4 presents selected examples illustrating common prediction patterns and error types observed across configurations.

Persian Text	**Expected**	**Prediction**	**Error Type**
این خبر مرا عصبانی کرد	Anger	Sad	Negative emotion confusion
از این فساد و دروغ خسته شدم	Disgust	Sad	Negative emotion confusion
هوا امروز معمولی بود	Neutral	Sad	Neutral class confusion
از امتحان فردا می‌ترسم	Fear	Fear	Correct after augmentation
باورم نمی‌شود تیم ما برنده شد!	Surprise	Happy	Positive emotion overlap

Table 4: Representative prediction examples and observed error patterns.

The qualitative results reveal systematic confusion among semantically related emotions, particularly within the negative affect spectrum. These patterns are consistent across multiple configurations and are further analyzed in the following discussion section.

7 Discussion

This section discusses the results presented in the previous section by interpreting observed performance patterns, relating them to prior work, and highlighting remaining challenges. The focus is on understanding the effects of semi-supervised scaling, class imbalance, and targeted data augmentation in the context of Persian (Dari) emotion classification.

7.1 Effectiveness of Semi-Supervised Dataset Expansion

The results demonstrate that semi-supervised learning provides a practical and effective mechanism for scaling emotion classification in low-resource settings. Expanding the dataset from a small manually annotated corpus to a large automatically labeled collection led to a substantial improvement in overall performance. This confirms that the representations learned during Phase 1 were sufficiently robust to serve as a labeling agent for large-scale data expansion.

At the same time, the results suggest that automatic labeling alone is not sufficient to address all challenges. While majority emotion classes benefited significantly from increased data volume, minority emotions did not improve proportionally. This observation is consistent with findings in prior work, which report that pseudo-labeling tends to reinforce dominant class patterns when label distributions are highly skewed.

7.2 Impact of Extreme Class Imbalance

A key finding of this study is the disproportionate impact of extreme class imbalance on macro-level performance. Although the eight-class model achieved high accuracy, the low F1 score for the *Fear* class substantially reduced the macro-F1 score. The seven-class ablation experiment provides further evidence that the instability of a single rare class can negatively affect aggregate evaluation metrics.

This behavior aligns with previous research emphasizing that accuracy alone is insufficient for evaluating fine-grained emotion classification under imbalance. The ablation results highlight the importance of explicitly analyzing rare classes rather than relying solely on global performance indicators.

7.3 Role of Targeted Data Augmentation

Reintroducing the *Fear* class through targeted data augmentation proved to be an effective strategy for mitigating extreme imbalance. Unlike generic augmentation approaches, the culturally informed generation of fear-related expressions led to substantial improvements in class-level performance without compromising accuracy on majority emotions.

These results support the argument that data-level interventions are often more effective than aggressive loss re-weighting in highly imbalanced emotion classification tasks. By focusing augmentation on a specific low-resource emotion and grounding it in realistic linguistic contexts, the model was able to better capture distinguishing features of fear-related expressions.

7.4 Error Patterns and Emotion Confusion

Despite the overall improvements, qualitative analysis reveals persistent confusion among semantically related emotions, particularly within the negative affect spectrum. Emotions such as sadness, anger, and disgust frequently overlap in lexical and contextual cues, making fine-grained discrimination challenging even with large training datasets.

Additionally, neutral statements were occasionally misclassified as emotional, suggesting that the boundary between neutral and low-intensity affect remains difficult to model. These patterns indicate that increased data volume alone does not fully resolve ambiguity inherent in emotional language and that structural or hierarchical modeling approaches may be required.

7.5 Limitations

Several limitations of this study should be acknowledged. First, automatic labeling introduces a degree of noise that cannot be entirely eliminated, even with careful preprocessing and validation. Second, the targeted augmentation strategy was applied only to a single emotion class; extending similar strategies to other low-frequency emotions may require additional linguistic analysis. Finally, the study focuses exclusively on text-based emotion classification and does not consider multimodal signals commonly present in social media content.

7.6 Summary of Insights

Overall, the discussion highlights three central insights: (1) semi-supervised learning is effective for scaling emotion classification in low-resource languages, (2) extreme class imbalance can significantly distort macro-level evaluation even when accuracy is high, and (3) targeted, culturally informed data augmentation offers a viable path toward improving minority emotion recognition. These insights provide a foundation for future refinement and extension of emotion classification systems for Persian (Dari).

8 Conclusion and Future Work

8.1 Conclusion

This report presented Phase 2 of an emotion classification study for Persian (Dari) social media text, building directly on the manually annotated dataset and baseline model developed in Phase 1. The primary objective of this phase was to address scalability and class imbalance through a semi-supervised learning framework combined with targeted data augmentation.

By leveraging the Phase 1 model for automatic labeling, the dataset was expanded from 4,000 manually annotated samples to approximately 390,000 samples. Experimental results demonstrate that this large-scale expansion substantially improved overall model performance. At the same time, the study revealed that extreme class imbalance—particularly for the *Fear* category—can significantly distort macro-level evaluation metrics, even when overall accuracy remains high.

The results further show that simple removal of rare classes improves aggregate performance but sacrifices emotion coverage. In contrast, reintroducing the *Fear* class through culturally informed data augmentation led to the best overall performance, improving both macro-F1 score and minority class recognition without degrading majority class accuracy. These findings confirm that data-level interventions, when carefully designed, are effective for mitigating imbalance in low-resource emotion classification tasks.

8.2 Future Work

While the proposed approach yields strong results, several directions remain open for future research. First, extending targeted augmentation strategies to other low-frequency emotions, such as *Disgust* or *Surprise*, may further improve overall balance and robustness. Second, hierarchical or multi-stage classification models could be explored to better separate semantically similar negative emotions and reduce confusion among sadness, anger, and fear.

Additionally, incorporating contextual or user-level information may help disambiguate subtle emotional cues that are difficult to capture at the sentence level alone. Finally, future work could extend the current text-based framework to multimodal emotion analysis by integrating visual or audio signals commonly present in social media platforms.

8.3 Final Remarks

Overall, my project demonstrates that semi-supervised learning combined with culturally grounded data augmentation offers a practical and effective solution for scaling emotion classification in low-resource languages such as Dari. Beyond the achieved performance improvements, the study provides methodological insights into handling extreme class imbalance and highlights the continued importance of qualitative analysis alongside quantitative evaluation. These contributions lay a solid foundation for future research and real-world applications of Persian emotion analysis systems.

9 Acknowledgments

I would like to express my sincere gratitude to my professor ao. Univ.-Prof. Mag. Mag. Dr.Werner Winiwarter for the continuous guidance and for sharing a broad range of relevant research papers related to my project. A selected subset of these works was incorporated into the study and proved instrumental in shaping the methodological choices and evaluation perspective adopted in my work.

References

[1] M. H. Farahani, H. Kazemi, *et al.*, "Parsbert: Transformer-based model for persian language understanding," *arXiv preprint arXiv:2005.12515*, 2020.

[2] X. Cui, "A comprehensive evaluation of transformer models for persian nlp tasks," *arXiv preprint arXiv:2507.11384*, 2025.

[3] Z. B. Farideh Majidi, "Macro-level evaluation metrics for emotion classification under class imbalance," *arXiv preprint arXiv:2507.11634*, 2025.

[4] M. G. . Łukasz Radliński1[0000 −0002 −7366 −3847] and J. K. . . −6896], "Fine-grained emotion classification across low-resource languages," *arXiv preprint arXiv:2507.14590*, 2025.

[5] S. F. X. Y. D. W. Y. Z. Shiyi Mu, Yongkang Liu, "Limitations of loss reweighting for imbalanced emotion datasets," *arXiv preprint arXiv:2507.14887*, 2025.

[6] S. K. A. Y. W. T. W. XIAOTIAN SU, NAIM ZIERAU, "Does data augmentation really help emotion classification?," *arXiv preprint arXiv:2507.21089*, 2025.

[7] S. K. D. M. . C. P. . Jiyu Chen1∗, Necva Bölücü1∗, "Targeted data augmentation for minority emotion classes," *arXiv preprint arXiv:2508.01161*, 2025.

[8] . D. N. D. H. Donya Rooein1 Flor Miriam Plaza-del Arco1, "Culturally grounded emotion modeling in conflict-affected regions," *arXiv preprint arXiv:2509.05719*, 2025.

[9] M. K. A. C. P. I. K. S. K. A. H. M. E. Bangzhao Shu1*, Isha Joshi1*, "Beyond accuracy: Qualitative evaluation of emotion classifiers," *arXiv preprint arXiv:2509.09593*, 2025.

[10] S. R. S. P. Kian Tohidi1, Kia Dashtipour2, "Error analysis and emotion boundary detection in transformer models," *arXiv preprint arXiv:2509.14922*, 2025.